

COSC 594

Review

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Announcement

Homework 4

Due on 11/21, 11:59 pm

Project Report

Due on 12/2, 11:59 pm

Class participation

#1 Scribing lectures

#2 Providing feedback

Two pillars

■ POLICY OPTIMIZATION

$$\underset{\pi}{\text{maximize}} \quad V^{\pi}$$

■ VARIATIONAL INFERENCE

$$\underset{p(x,z)}{\text{maximize}} \quad \text{ELBO}(p(x,z))$$

Part I

RL for GenAI

(generative language models and diffusion models)

Summary of Part I

■ ALIGNMENT AND COMPOSITION OF LANGUAGE MODELS

reward-based (mult-objective, constrained), preference-based
model merging (regularization, constraints)

■ ALIGNMENT AND COMPOSITION OF DIFFUSION MODELS

reward-based (path-/point-wise KL), preference-based, inference-time
conjunction, disjunction, and negation (regularization, constraints)

■ UNLEARNING IN LANGUAGE AND DIFFUSION MODELS

negation (regularization, constraints)

Part II

GenAI for RL

(policy optimization)

Summary of Part II

■ DIFFUSION MODELS FOR POLICY OPTIMIZATION

offline RL, online RL

■ DIFFUSION MODELS FOR ROBOTICS

motion planning (single-/multi-agent)

■ OTHER TOPICS

LLMs as world/reward models

Challenges

■ RL for GenAI

How to use RL to improve GenAI by optimizing for user requirements?

- ★ controllable generation
- ★ scalable sampling algorithms
- ★ optimality and efficiency characterization

■ GenAI for RL

How to use GenAI to reduce RL sample complexity and safety risks?

- ★ world information
- ★ safe exploration
- ★ scalable learning algorithms
- ★ optimality and efficiency characterization

Vision

Imagination (GenAI)

+

Search (RL)



Figure AI

Feedback

**Any suggestions on lectures, homeworks, and projects
are welcome.**

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